

CS5188 Virtual Reality Technologies and Applications

Course Project Specification

1. Objectives

The aims of this course project are for students to have some hands-on experience in VR/AR programming and to develop a VR/AR application. Your objective in this project is to complete a few basic tasks and then develop either a VR application or an AR application.

2. Requirements of the Course Project

This course project can be carried out as an individual or a group project. The maximum number of members in each group is 3. However, we would expect you to do more work if you have more people in your group. The responsibility of each group member should be clearly indicated in the report, and it is possible that different members of a group will receive different marks.

You are given a VR demo program and an AR demo program. Both of them are based on the Unreal Engine 5 (supporting C++ and Blueprint). You can select one of them (VR or AR) to develop your course project. Based on your selected demo program, you will need to extend it to include more features (the basic requirements) and then develop it into an application (the advanced requirement). Note that you **MUST** build your work on the selected demo program. However, you should not take the game provided in the demo program as your application, but develop your own application.

The course project is divided into two parts. The **basic requirements** are designed for you to practice some basic VR/AR programming skills (more like a warm-up exercise). The **advanced requirement** is for you to develop a real application, and is more flexible in terms of what you can do. The basic requirements and advanced requirement account for 45% and 60%, respectively, of the total mark for this project. The extra 5% is a bonus, but the final mark will be clipped to 100.

2.1 Basic Requirements (45%)

You are required to choose three (of a maximum of 15 marks each) of the following items for the basic requirements. The items that you choose should be used in the Advanced Requirement. The mark that you will receive for each item will depend on the quality of your work and the amount of effort that you put into it.

(1) ***Creating a new object***

Create and add a new object into the scene. This object **MUST** be created by you. You can use a tool, e.g., 3D Studio Max or Google Sketchup, to help you do so and then import it to your scene.

(2) ***Object manipulation***

Enable the use of the physical keyboard/mouse to manipulate the properties (size, position and angle) of an object. This can be done through binding an object to the keyboard/mouse.

(3) ***Audio ambience***

Incorporate a sound effect to an object such that the object will produce some sound when a certain event occurs, e.g., if the object is being pressed. It may also change the sound volume when another event occurs, e.g., if the user is moving near or further away from it.

(4) ***Adding autonomous objects***

Allow an object to move around automatically and react to some user actions. (For example, an autonomous object may fall down when shot at.)

(5) ***Pop-up menu***

Add a pop-up menu to change the environmental property, e.g., to switch between different lighting conditions (ambient, diffuse, specular and position).

2.2 Advanced Requirement (60%)

You are expected to extend the program into a VR/AR application to solve a real world problem, such as helping people with fear of heights (<https://www.youtube.com/watch?v=2F4XIQWL75c>) or helping people learn English (<https://www.youtube.com/watch?v=idCOgNR8U-g>). You will receive more marks with a novel and/or useful application. As the advanced requirement contributes to a significant portion of the total mark, you are suggested to put more effort in this part. We will grade this part of the work on the following factors: (1) the usefulness/novelty of your application; (2) the amount of work; (3) whether you are making use of the work done in the Basic Requirements to form your application; and (4) your final presentation. Note that we will need to make use of the work done in the Basic Requirements in here. Otherwise, a significant portion of your marks will be deducted from this part.

For your information, in previous years, a lot of students would write a game for this part of the project. However, most of these games are very simple, and these students ended up receiving low marks for this part. So, unless you can really come up with a different idea for the game, I would not suggest you to write a game as the application here.

3. Submission Details

The submission deadline is: ***April 29, 2026.***

Each group needs to submit the following items as a zip file on Canvas. If your zip file is too large, you may submit a link where we can download your zip file. However, you should keep the file size of your zip file to no more than 1GB in total. If you do need to have a very large file size, you will need to provide a justification with your submission.

/Project:

All files used in this project in the Unreal Engine Project directory. A README file should be included to tell us how to run your project. You will need to include both source codes and executable release. Please remove any redundant/unused resources from your submission. This can greatly reduce your final file size.

/Video:

A demo video that guides the marker through the main contributions of the work.

/Report:

A report to describe what you have done. For the basic requirements, it should describe the three items that you have done. You should provide sufficient information for us to know what you have done and your amount of effort. For the advanced requirement, it should describe your application. Again, you should provide sufficient information for us to know what your application is. If there are some new ideas, you should discuss them. You should also let us know where your efforts are in developing the project.

Note that we will not be grading the report itself. Hence, there is no need to submit a large report. It can just be a few pages providing the following information:

- (1) A cover page that indicates your name(s) and student ID(s).
- (2) You need to declare the responsibilities of each group member in the project, including: programming tasks, report writing, and group coordination. You will need to declare this twice, in the first submission and in the second submission.
- (3) A section to briefly discuss each of the three items that you have done for the basic requirements. For each item, please point out the objective of your work and the final effects produced, with screen captures of the results. If there are particular efforts that you have put into an item, you can discuss them so that we will pay particular attention to them.

Note that your submission must contain the above three items (i.e., /Program, /Video, /Report). Marks will be deducted if any is missing. It is your responsibility to make sure that your submitted program works. Note also that your work must be built based on the pre-setting environment.